

Design Overview

The design can be divided into two main flows:

1. **Onboarding / Registration of Routable Agents** — Store routers and their routable destinations in the relational database. # control tower
2. **Routing Inference** — Load the destinations available for the current router and provide them to the voice agent for route selection.

1. Onboarding / Registration of Routable Agents

All router and routable-agent configuration is stored directly in PostgreSQL.

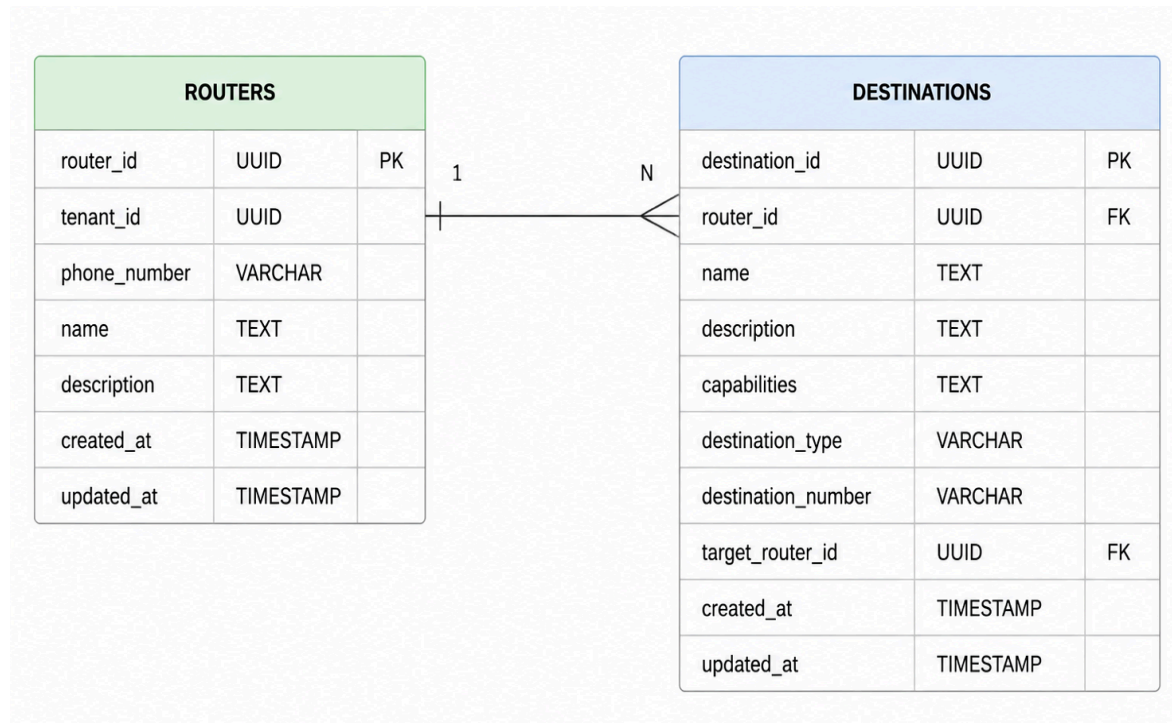
The database acts as the **single source of truth**. No RAG pipeline, embeddings, vector search, or chunking is required.

A routable destination can represent:

- Voice Agent
- Human-Agent Team
- Another Router



Database Structure



router_id determines which destinations belong to a router.

target_router_id is used when a destination points to another router.

JSON SCHEMA FOR THE SYSTEM PROMPT

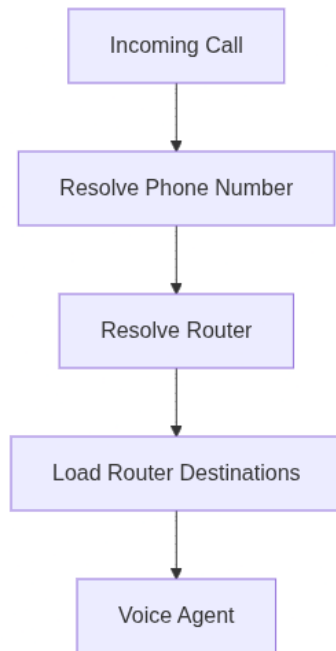
```
{
  "router_id": "router_123",
  "phone_number": "+1-xxx-xxx-1234",
  "name": "Main Router",
  "description": "Routes incoming customer calls",
  "destinations": [
    {
      "destination_id": "dest_001",
      "name": "Billing",
      "description": "Handles billing and payment related queries",
      "capabilities": "Refunds, duplicate charges, payment issues, invoices",
      "destination_type": "agent",
```

```
"destination_number": "+1-xxx-xxx-5678",  
  
"target_router_id": null  
  
},  
  
{  
  
"destination_id": "dest_002",  
  
"name": "HR Router",  
  
"description": "Routes employee and HR related queries",  
  
"capabilities": "Salary, leave, benefits",  
  
"destination_type": "router",  
  
"destination_number": null,  
  
"target_router_id": "router_456"  
  
}  
  
]  
  
}
```

2. Routing Inference

When a call arrives, the dialed phone number is used to identify the corresponding router.

The application then loads all destinations configured for that router.



For example:

Main Router

Available Destinations:

- Billing
- Sales
- Technical Support
- HR

The voice agent receives these destinations together with their descriptions and capabilities.

3. Route Selection

The voice agent uses the caller's conversation and the available destinations to select the most appropriate route.

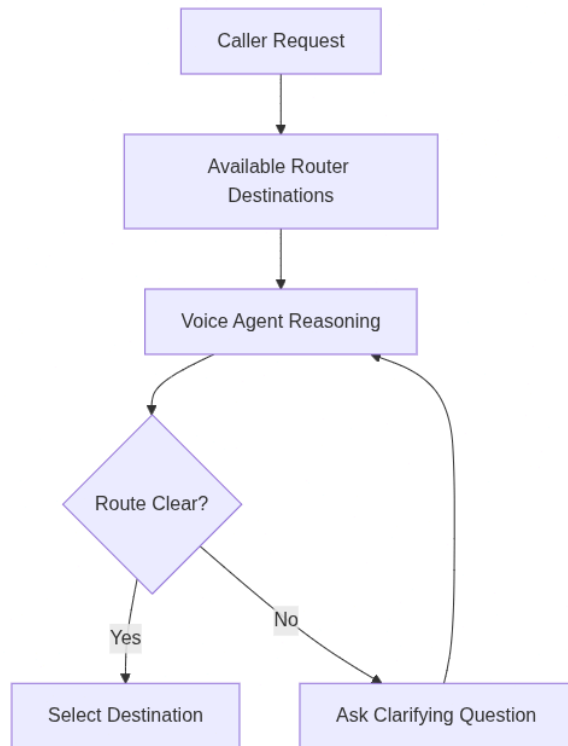
For example:

Caller: "I was charged twice and want a refund."

Available Destinations:

- Billing
- Sales
- Technical Support

Selected: Billing



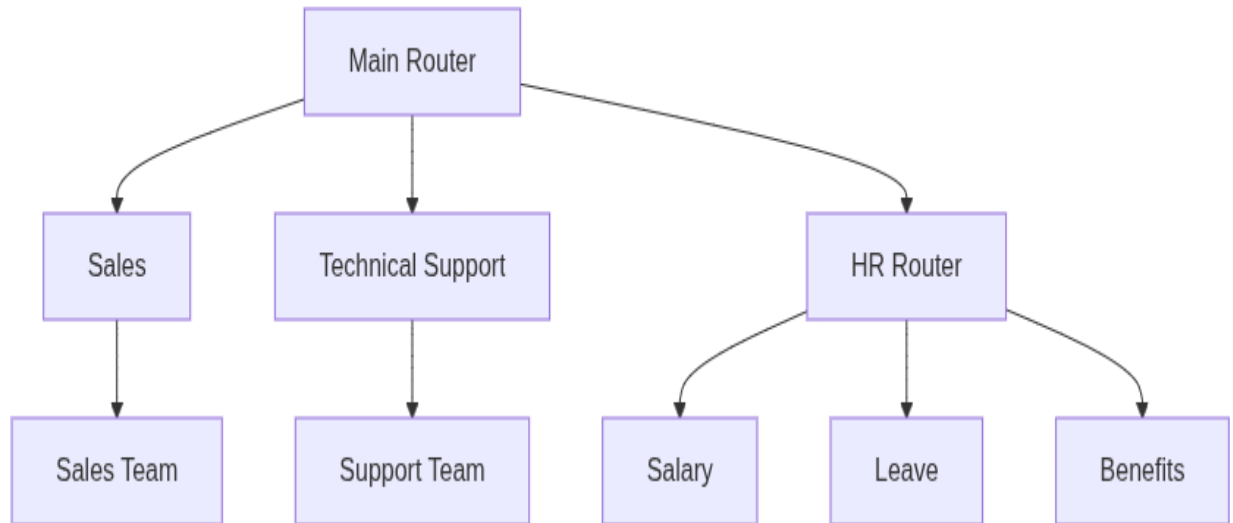
If the request is ambiguous, the voice agent can ask the caller for additional information before selecting a destination.

4. Execute Route

Once a destination is selected, the application forwards the use by making call to twilio or in same call prompt switching by calling the data api

5. Hierarchical Routing

Routers can contain destinations that point to other routers.



Only the destinations belonging to the **current router** are loaded.

For example:

Main Router



Voice Agent selects HR



Load HR Router



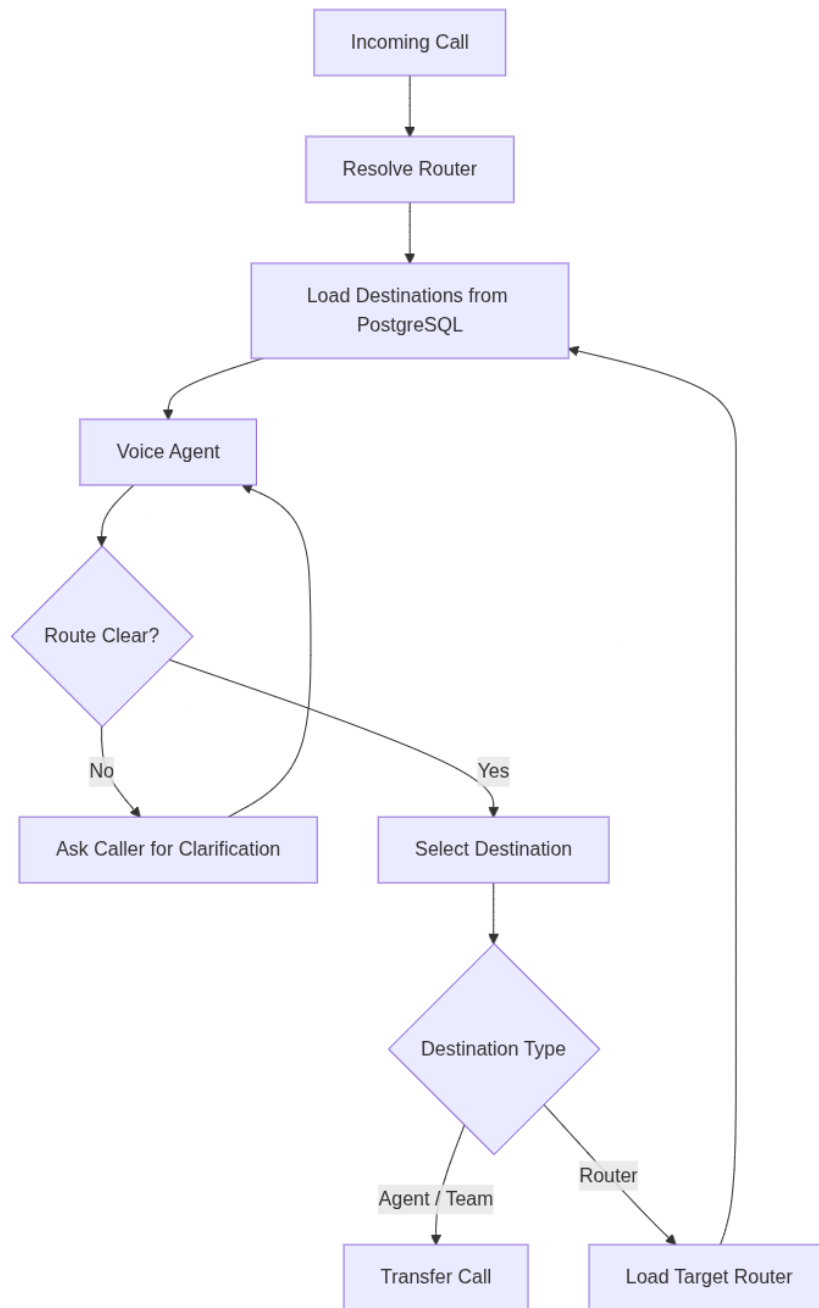
Salary / Leave / Benefits



Voice Agent selects Benefits

This prevents the entire routing hierarchy from being loaded into the model at once.

6. Complete Routing Flow



Limitations

- **Large routers** — Too many destinations can increase the model context and latency.
- **Overlapping destinations** — Similar descriptions may make route selection difficult.
- **Ambiguous requests** — The voice agent may need to ask clarification questions.
- **Incorrect routing** — A fallback destination should be available when the voice agent cannot confidently determine a route.

